DATA MINING

Probabilities

Measuring how likely an event really is

0 impossible

certain 1

WHY IT MATTERS

Decisions are made under uncertainty

Life is full of uncertain events, and we usually have to weigh the possible outcomes before we decide. Executives face the same problem at a much larger scale, whether they are funding a research project, approving a merger, or walking away from a buyout.

Probability and statistical data let them estimate how likely each outcome is, then make the call with evidence instead of instinct.

? **What is the chance of success?**

? **What is the probability that we fail?**

? **Is the risk actually worth taking?**

These are probability questions, even when nobody calls them that.

DEFINITION

So what is probability?

1 In plain words

The chance of something happening.

2 More academically

The likelihood of an event occurring.

Same idea, two registers. Use the second one in your write-ups.

The word event has a specific meaning here, and that is the next thing we need to pin down.

TERMINOLOGY

An event is an outcome, or a set of them

An event is a specific outcome, or a combination of several outcomes. The outcomes themselves can be almost anything.

A

Getting heads

when flipping a coin

B

Rolling a four

on a six sided die

C

Running a mile

in under six minutes

One experiment can hold several events. A coin flip has two possible outcomes, so it has two events, and each one needs its own probability.

NOTATION

We write probabilities as numbers

Knowing an event is merely likely or unlikely is rarely enough. We want to measure and compare, so we express probability numerically.

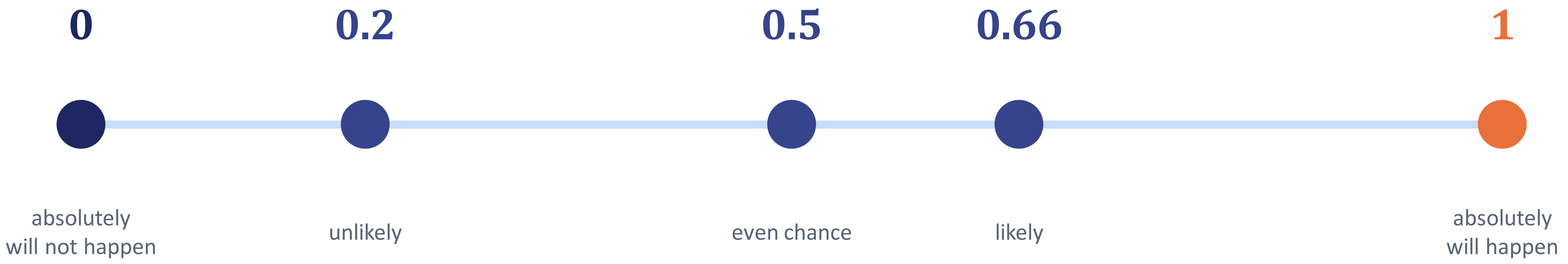
20%	1/5	0.2
Percentage	Fraction	Real number ***conventional***

By convention we use real numbers between 0 and 1, so 0.2 rather than 20% or one fifth.

INTERPRETATION

Reading a probability value

Higher value, higher likelihood. Nothing more mysterious than that.



Most events we care about land somewhere in the middle, so values like 0.2, 0.5 and 0.66 are what you should expect to see.

THE CORE RULE

How we actually compute it

$$P(A) = \frac{\text{number of preferred outcomes}}{\text{total number of possible outcomes}}$$

1 Preferred outcomes

The results we want to see happen. Some textbooks call these favorable outcomes. Both terms mean the same thing.

2 Sample space

Every outcome that is possible at all. This is the denominator, and getting it right is usually the hard part.

WORKED EXAMPLES

Two straightforward cases

Flipping a coin and getting heads

1 ÷ 2

preferred outcomes

possible outcomes

$P(\text{heads}) = 1/2 = 0.5$

Heads is the only preferred outcome, and there are two possible outcomes in total.

Rolling a four on a six sided die

1 ÷ 6

preferred outcomes

possible outcomes

$P(\text{four}) = 1/6 \approx 0.167$

Still one preferred outcome, but the sample space is now three times larger.

COMBINED OUTCOMES

When an event covers more than one outcome

Rolling a number divisible by three means we would accept either a three or a six.

1	2	3	4	5	6	Sample space stays at 6
		preferred			preferred	

Preferred outcomes	2	*the faces 3 and 6*
Possible outcomes	6	*the die still has six sides*
Probability	$2/6 \approx 0.33$	*roughly a one in three chance*

A USEFUL SHORTCUT

Independent events multiply

The probability of two independent events happening at the same time equals the product of their individual probabilities.

P(ace of spades)	=	P(ace)	×	P(spade)
1/52		1/13		1/4

! We will define independence properly in a later lecture. For now, treat it as events that do not affect each other.

SCALE CHECK

The same rule, at an absurd scale

Winning the lottery follows the identical logic. Preferred outcomes are the tickets you bought. Possible outcomes number upward of 175 million.

0.0000000057

One ticket, roughly a 1 in 175,000,000 chance

0.5

A single coin toss, for comparison

Worth asking yourself: how much do your odds really improve if you buy two tickets? Or five?

Why do we express probabilities numerically?

- **A** To determine whether they are likely or unlikely.
- **B** To compute which event is relatively more likely.
- **C** We don't express probabilities numerically.
- **D** To be able to feed them as data to computers.

Which of these is our preferred representation of the probability of winning a game of rock-paper-scissors on the first go?

Hint Use the favored over all formula.

A

33.(3)%

B

1/3

C

0.3(3)

D

1 to 2

What is the probability of drawing a spade from a standard deck of playing cards?

Hint A standard deck consists of 52 playing cards, 13 of which are spades.

A	B	C	D
0.5	0.25	0.2	0.08

PART TWO

Expected values

What we expect the outcome to be if we run an experiment many times

First, what is an experiment?

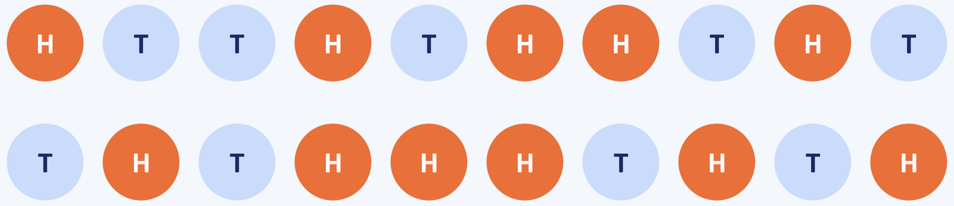
1 Trial

One flip of the coin, with the outcome written down. That single action is a trial.

2 Experiment

Many trials run together. Toss a coin 20 times and record all 20 outcomes, and that whole process is one experiment.

One experiment, 20 trials



11 heads out of 20 trials, so the experimental probability here is 0.55. Close to the theoretical 0.5, but not identical.

TWO KINDS OF PROBABILITY

Where the number comes from

T

Theoretical

Also called the true probability. Worked out from the structure of the problem itself, the way we did with coins, dice and cards.

E

Experimental

Worked out by actually running the trials and counting what happened. No structural reasoning required.

The two are rarely identical, but the experimental value is a good approximation. We reach for it whenever the true probability is unknown, or would take far too many factors to compute.

Same shape as the formula you already know

$$P = \frac{\text{number of successful trials}}{\text{total number of trials}}$$

Eight out of ten visits to my local shop, I have to wait in line.

80%

there is a queue

20%

there is no queue

Calculating the true probability here would mean accounting for far too many factors. The experimental one takes a minute and is genuinely useful.

DEFINITION

The expected value

$E(A)$

The outcome we expect to occur when we run an experiment.

Which formula you use depends on what kind of outcome you are dealing with.

Categorical outcomes

Suits, colours, yes or no

$E(A) = P(A) \times n$

Numerical outcomes

Points, prices, quantities

Add up value × probability

Multiply the probability by the trials

$$E(A) = P(A) \times n$$

How many spades should we expect if we draw a card 20 times, recording it and returning it to the deck each time?

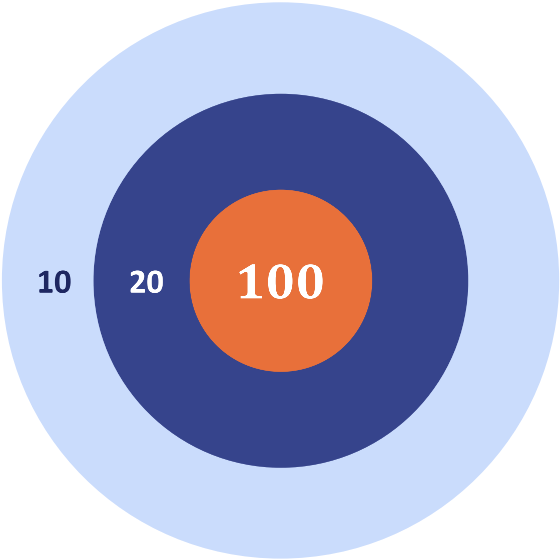
0.25	20	5
P(spade)	trials	E(spade)

An expected value of 5 does not guarantee 5. You could draw 4 spades, or 6, or in principle all 20.

NUMERICAL OUTCOMES

Weight every value by its probability

Take each value in the sample space, multiply it by its probability, then add them all up. You always hit this target, but not always the middle.



LAYER	POINTS	P	VALUE
Outer ring	10	0.5	5
Second ring	20	0.4	8
Bullseye	100	0.1	10

Expected value 23

WHY IT MATTERS

You can never score 23 with one shot

So why bother? Because expected values let us make predictions about the future based on past data.

Predictions usually come as intervals

The future carries too much uncertainty for a single exact number, so we forecast a likely range instead. Meteorologists do this constantly.

Expect between three and five feet of snow tomorrow morning.

Temperatures rising up to 90 degrees on Wednesday.

Up next: making reasonable predictions with the probability frequency distribution.

Which of the following classifies as an experiment but not a trial?

- **A** Flipping a coin 20 times and recording the 20 different outcomes.
- **B** Flipping a coin 20 times and recording the outcome of the last throw.
- **C** Flipping a coin once and recording the outcome.
- **D** Flipping a coin 20 times without recording any of the outcomes.

Why do we use experimental probabilities?

- **A** Because they are good predictors for theoretical ones.
- **B** Because they are easy to compute.
- **C** Because they are easy to compute and serve as good predictors for theoretical ones.
- **D** None of the above.

Why do we use intervals when forecasting future events?

- **A** Because the expected value might have a low probability of occurring.
- **B** Because we want to increase the likelihood of our predictions being accurate.
- **C** Because the expected value could be unattainable.
- **D** All of the above.

PART THREE

Probability frequency distribution

The probability of every possible outcome, laid out side by side

THE FAMOUS EXAMPLE

Throwing two dice and adding the tops

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

36 possible outcomes

Six results on the first die, and six on the second whatever the first one was.

Read every cell as the sum of its row and column. Repeating entries run along the secondary diagonal and every diagonal parallel to it. Seven fills that whole diagonal, so it appears six times.

$$P(\text{sum} = 7) = 6/36 = 1/6 \approx 0.167$$

THE CATCH

The expected value is seven. So what?

How we get there

These are numerical outcomes, so we use the archery formula. Give every unique sum a probability based on how often it appears in the table, multiply, then add it all up.

E(sum)

7

Most probable is not the same as likely

Seven carries a probability of only 1/6, so you cannot reasonably bet on the sum landing exactly there.

And how would you even know?

Nothing so far proves seven is the most probable sum. A frequency distribution is what settles it.

DEFINITION

Probability frequency distribution

A collection of the probabilities for each possible outcome.

It is normally expressed one of two ways, and we will build both from the dice table.

1 As a table

Every outcome listed in ascending order, with its probability beside it.

2 As a graph

The same information as bars, which makes the shape of the spread obvious at a glance.

STEP ONE

Count how often each sum appears

Work through the sample space table and record, for every unique sum, the number of times it features. That count is the frequency of the outcome.

Sum	2	3	4	5	6	7	8	9	10	11	12
Frequency	1	2	3	4	5	6	5	4	3	2	1

A sum of eight turns up in five different cells, so eight has a frequency of five. Listing every outcome in ascending order with its frequency gives us a frequency distribution table.

STEP TWO

Turn each frequency into a probability

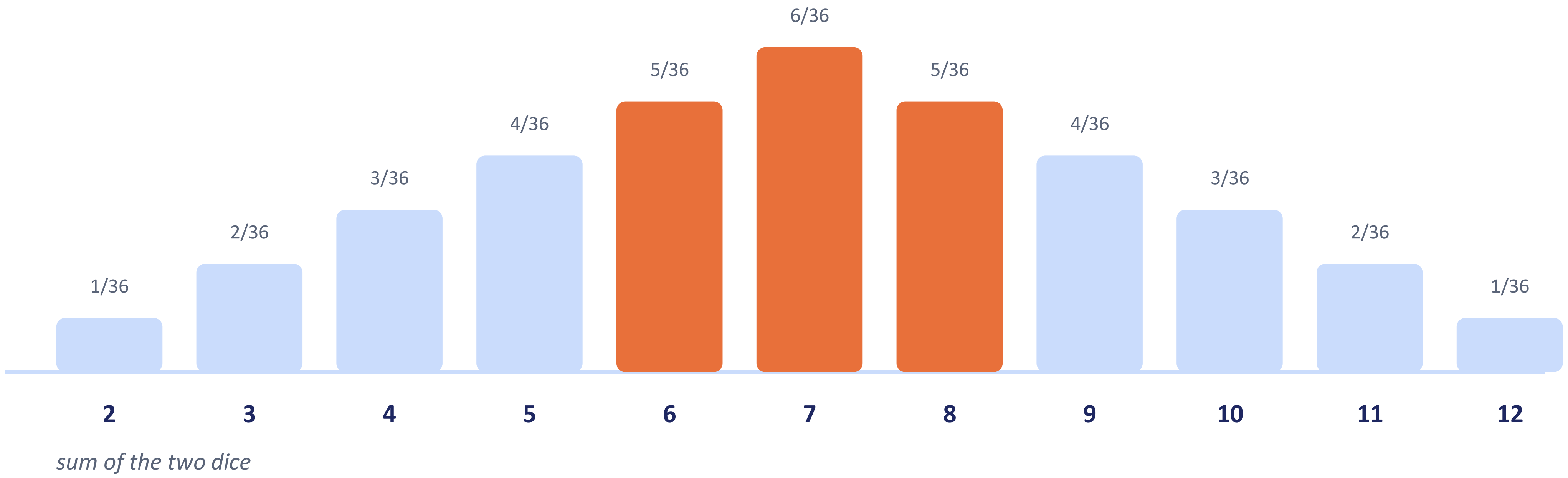
Divide the frequency of every outcome by the size of the sample space. With 36 possible outcomes, that gives the true probability of each sum.

Sum	2	3	4	5	6	7	8	9	10	11	12
Frequency	1	2	3	4	5	6	5	4	3	2	1
Probability	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

That full set of probabilities is the probability frequency distribution.

AS A GRAPH

Build your interval around the expected value



The tallest bars cluster around the expected value, so an interval built there carries the highest probability.

Up next: the complement, the opposite of an event.

Going back to the example with the 2 standard six-sided dice, why is the probability of rolling a sum of 7 equal to one sixth?

A A standard die has exactly 6 sides, so the probability of the expected value has to be one sixth.

B There are 36 favorable outcomes and 6 outcomes in the sample space, so the favorable over all formula results in 6/36, or simply 1 over 6.

C There are 6 favorable outcomes and 36 outcomes in the sample space, so the favorable over all formula results in 6/36, or simply 1 over 6.

D None of the above.

What does the frequency of a value within the sample space represent?

- **A** The number of times the value features in the sample space.
- **B** The likelihood of the outcome occurring.
- **C** A collection of probabilities for each possible outcome.
- **D** A table or graph matching how often each value occurs in the sample space.

Going back to the dice example and the 6 by 6 table, which sum of the two dice has a frequency of 3?

A

10

B

4

C

4 and 10

D

3

PART FOUR

Complements

The opposite of an event, and why it saves so much work

DEFINITION

Everything the event is not

The complement of an event covers whatever the event leaves out. As the name suggests, it completes the rest of the sample space.

1 2 3 4 5 6

A

Rolling an even number

2, 4 or 6

A'

Rolling an odd number

1, 3 or 5

Every event has one. We write the complement of A as A', and the complement of a complement is the event itself, so (A')' = A.

THE RULE UNDERNEATH IT

All the outcomes together add up to one

Toss a coin and you are guaranteed heads or tails. Account for both and you have exhausted the sample space, so you are 100% certain of landing on one of them.

0.5	+	**0.5**	=	**1**
heads		tails		certainty

A probability of one is the same as being absolutely certain. Since one of the two outcomes has to happen, their probabilities have to total one. This holds for any event and its complement.

DIAGNOSTICS

If your probabilities do not total one

Both cases tell you something specific has gone wrong.

> 1

Sum is greater than one

You cannot be surer than absolutely sure, so 1.5 makes no intuitive sense. This happens when some of the assumed outcomes can occur at the same time, which means you are double counting. Bayesian notation deals with this later.

< 1

Sum is less than one

You have not accounted for one or more possible outcomes. Anything below one is not guaranteed to occur, so there must be a piece of the sample space still missing.

THE FORMULA

Subtract from one and you are done

$$P(A') = 1 - P(A)$$

1

Because everything totals one

If the event and its complement together fill the sample space, then whatever is not in the event must make up the rest.

2

Reach for it when the event is crowded

The more outcomes satisfy what you want, the more work direct counting becomes. Flip to the complement and count the few outcomes you do not want instead.

WORKED EXAMPLE

Rolling a 1, 2, 4, 5 or 6

Two routes to the same number. One of them is considerably shorter.

The long way

Add the probability of each outcome you want. Every face carries a probability of 1/6, and there are five of them.

$$1/6 + 1/6 + 1/6 + 1/6 + 1/6$$

$$= 5/6$$

The complement way

Getting a 1, 2, 4, 5 or 6 is just another way of saying not getting a three. So subtract.

$$1 - P(3) = 1 - 1/6$$

$$= 5/6$$

Same answer, one subtraction instead of five additions. On bigger problems the gap is far wider.

WHERE THIS GOES NEXT

Back to the lottery

With the basic notions in place, we return to the lottery example from the very first lesson and open up the field of combinatorics.

1

Variations

Order matters, and repetition may be allowed

2

Permutations

Arranging a set of elements among themselves

3

Combinations

Selecting a group where order is irrelevant

What each term means, when to use it, and how to compute it.

Which of the following statements is FALSE?

A The complement of an event is everything the event is not.

B The event and its complement add up the entire sample space.

C The complement of “Heads” is “Tails”.

D The complement of rolling a 6 is rolling a 1.

What is the complement of NOT rolling a 3?

- **A** Not rolling a 3.
- **B** Rolling a 3.
- **C** Not rolling a 1, 2, 4 or 5.
- **D** Rolling a 4.

Which of these are complements?

- **A** Positive and negative numbers.
- **B** Non-negative and non-positive numbers.
- **C** Positive numbers and non-negative numbers.
- **D** Negative numbers and non-negative numbers.

DATA MINING · PROBABILITIES

End of presentation

Thanks for following along.